

01 — Departure from CSR: $\tilde{\delta}$ and null tables

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1 Theory

Setting

- Window $W = [0, 1]^2$, fixed
- $\bar{n}$: expected number of points (deterministic model parameter)
- Pattern $x = \{x_1, \dots, x_n\} \subset W$; $n = |x|$ realised number of points (random around $\bar{n}$)
 - Poisson, Matérn II: $n/\bar{n} \in [0.90, 1.11]$ (5–95%)
 - Thomas, nested, LGCP: $n/\bar{n} \in [0.77, 1.25]$ (overdispersed, clusters arrive whole)

Estimator

- Ripley's K , isotropic edge correction, λ^2 estimated by $n(n-1)$:

$$\hat{K}_x(r) = \frac{1}{n(n-1)} \sum_{i \neq j} w(x_i, d_{ij}) \mathbf{1}\{d_{ij} \leq r\}, \quad w(x, d) = \frac{2\pi}{\text{length of } \partial b(x, d) \cap W}$$

- Centred L -function of a pattern x , and its model counterpart:

$$\hat{\phi}_x(r) = \hat{L}_x(r) - r = \sqrt{\hat{K}_x(r)/\pi} - r, \quad \phi(r) = L(r) - r \quad (= 0 \text{ under CSR})$$

- Evaluated on the radius grid: $\hat{\phi}_x = (\hat{\phi}_x(r_1), \dots, \hat{\phi}_x(r_{512})) \in \mathbb{R}^{512}$
- Radii $r_k = 0.25 k/512$, $k = 1, \dots, 512$ ($r = 0$ dropped: identically zero)

Radius set

Everything in this subsection is defined for fixed n . The test and $\tilde{\delta}$ only use radii in $R(n)$.

- At small r , $\hat{\phi}_x(r)$ is a function of a small count: the pairs of x closer than r . Under $\mathbb{P}_{0,n}$ we expect

$$\mu(r; n) = \frac{1}{2}n(n-1)\pi r^2$$

such pairs (edge-corrected)

- The maximum in M_n (next subsection) treats all radii alike only if each standardised deviation $(\hat{\phi}_X(r) - m_0)/s_0$ has roughly the same null distribution, i.e. $\approx N(0, 1)$
- Few expected pairs $\Rightarrow \hat{\phi}$ takes a handful of values; standardised deviation discrete and skewed, not $N(0, 1)$. Checked on the null curves at $n = 99$:
 - $\mu < 0.3$: heavy right tail, $\mathbb{P}(|z| > 3)$ $10\times$ Gaussian
 - $\mu \in [3, 10]$: still $3\times$ Gaussian
 - $\mu \geq 10$: Gaussian tail rate, skewness and kurtosis
- Heavy-tailed radii win the null maximum too often $\Rightarrow c_{95}$ inflated by counting noise $\Rightarrow$ power lost at every other radius (c_{95} at $n = 99$: 3.07 with $p = 10$, 3.43 with $p = 0.1$)

- Keep radii with at least p expected pairs, $p = 10$ (usual threshold for the normal approximation to a Poisson count):

$$R(n) = \{r_k : \mu(r_k; n) \geq p\} = [r_{\min}(n), 0.25], \quad r_{\min}(n) = \sqrt{\frac{2p}{\pi n(n-1)}} \approx \frac{2.52}{n}$$

- $r_{\min}$: 0.13 ($n = 20$), 0.025 ($n = 100$), 0.0034 ($n = 800$), 0.0015 ($n = 2000$). More points $\Rightarrow R(n)$ reaches down to finer scales

Null distribution

Everything in this subsection is defined for fixed n . In practice, we compute the null distribution at every n on a grid.

- Null $\mathbb{P}_{0,n}$: n i.i.d. uniform points on W (CSR conditioned on n); $\mathbb{E}_{0,n}$, $\text{sd}_{0,n}$ under it
- Generate 12 000 CSR clouds from $\mathbb{P}_{0,n}$
- Estimate null moments for $r \in R(n)$ using half of the CSR clouds:

$$m_0(r; n) = \mathbb{E}_{0,n}[\hat{\phi}_X(r)], \quad s_0(r; n) = \text{sd}_{0,n}[\hat{\phi}_X(r)]$$

- Using remaining half of CSR clouds, compute maximum deviation of each cloud x :

$$M_n(x) = \max_{r \in R(n)} \frac{|\hat{\phi}_x(r) - m_0(r; n)|}{s_0(r; n)}$$

- Determine critical value: $c_{95}(n) = 95\%$ quantile of M_n over these clouds (estimates the 95% quantile of $M_n(X)$, $X \sim \mathbb{P}_{0,n}$)

Test (studentised global envelope; Myllymäki et al. 2017)

$$S(x) = \frac{M_n(x)}{c_{95}(n)}, \quad n = |x|$$

- Under the null, $\mathbb{P}_{0,n}(S > 1) = 0.05$ by definition of c_{95} (up to Monte Carlo error)
- Reject CSR at 5% iff $S > 1$

Distance $\tilde{\delta}$

$$\tilde{\delta}(\theta, \bar{n}) = \frac{1}{c_{95}(\bar{n})} \max_{r \in R(\bar{n})} \frac{|L_\theta(r) - r|}{s_0(r; \bar{n})}$$

- Same functional as S , applied to the closed-form L_θ at $\bar{n}$, uncentred ($m_0 = 0$)
- $\tilde{\delta}(\text{CSR}) = 0$ exactly
- $\tilde{\delta} = 1$: true curve on the rejection boundary $\Rightarrow$ power well above 5%; measured 0.64–0.78 (02), not 50%, because pattern noise adds to the maximum
- Units: null s.d.s, relative to the critical value
- Property of the model, known before simulation $\Rightarrow$ usable as a sweep coordinate
- At fixed shape, $\tilde{\delta}$ grows with $\bar{n}$ (s_0 shrinks)

$\tilde{\delta}$ does not transfer well for Matérn II

- Hard core R : no two points closer than $R \Rightarrow L(r) - r = -r$ for $r < R$
- Size of the hard core in pairs: the CSR pairs it forbids,

$$\mu_R = \mu(R; \bar{n}) = \frac{1}{2} \bar{n} (\bar{n} - 1) \pi R^2 = p \left(R / r_{\min}(\bar{n}) \right)^2$$

- For $\tilde{\delta} \lesssim 1$ the hard core lies below $r_{\min}$ ($\mu_R < p$), where $\tilde{\delta}$ does not look. $\tilde{\delta}$ sees only the deficit it leaves above $r_{\min}$, which decays like $1/r \Rightarrow$ maximum at $r_{\min}$ for 82% of Matérn $\theta \Rightarrow$ Matérn $\tilde{\delta}$ is set by the cutoff p , not by the model
- A nearest-neighbour test sees the hard core directly: reject CSR if no pair is closer than d , with $\mu(d) \approx 3$ ($\mathbb{P}_0(\text{no pair within } d) \approx e^{-\mu(d)} = 0.05$; 2.97 measured)
- Same $\tilde{\delta}$, very different difficulty (all 20 000 simulated Matérn patterns; nearest-neighbour test calibrated to 5% on the Poisson pool):

$\tilde{\delta}$ bin	(0.1, 0.2]	(0.2, 0.3]	(0.3, 0.5]	(0.5, 1]	(1, 2]	(2, 4]
L test power	0.06	0.07	0.12	0.34	0.86	1.00
Nearest-neighbour test power	0.23	0.63	0.97	1.00	1.00	1.00

- So $\tilde{\delta}$ is a fair difficulty axis only for departures visible inside $R(\bar{n})$ (the cluster families). For Matérn we use μ_R as the regime coordinate:
 - Unit: pairs removed; detectable by the nearest-neighbour test from $\mu_R \approx 3$ on
 - Same ordering as $\tilde{\delta}$ (Spearman 0.999) and free of $\bar{n}$: at fixed μ_R , L -test power flat across $\bar{n}$ terciles while R varies $4\times$
 - Sweep unchanged: Matérn still drawn on $\tilde{\delta} \in [0.1, 4]$, which induces $\mu_R \in [1, 46]$
 - Matérn not pooled with the cluster families in $\tilde{\delta}$ -binned results

Smoothing in n

- True m_0 , s_0 , c_{95} smooth in n ; raw per- n estimates noisy
- Dense grid alone does not help: leave-one-out interpolation error on the old 20-point grid $\approx$ Monte Carlo noise
- Fit across $n \Rightarrow$ borrow strength from neighbouring n ; defined at every n , no interpolation rule
- $s_0 \propto n^{-1}$ at small r , slower at large $r \Rightarrow$ fit $n m_0$, $n s_0$ (slowly varying)
- Cubic B-splines in $\log n$

2 Implemented pipeline

Code

src/cloudforger/classical/lfunction.py	$\hat{K}$, $\hat{L} - r$, weights, $L - r$ from $K - \pi r^2$
src/cloudforger/departure/config.py	config, n grid, paths
src/cloudforger/departure/simulate.py	null patterns $\rightarrow$ stored curves
src/cloudforger/departure/fit.py	curves $\rightarrow$ spline tables
src/cloudforger/departure/tables.py	tables $\rightarrow S$, $\tilde{\delta}$
src/cloudforger/departure/validate.py	size on fresh patterns
scripts/departure.py	CLI: simulate, fit, validate
configs/departure/	config.yaml, tables.npz, report.json

Configuration

- n grid: $\text{round}(20 \cdot 1.03^k)$ up to 2000 $\Rightarrow$ 153 values

- Per grid n : 6000 fit + 6000 calibration patterns
- Validation: 50 000 patterns, n log-uniform integer on $[20, 2000]$
- $p = 10$ pairs; $\alpha = 0.05$; knots: 8 interior (moments), 4 interior (c_{95})

Steps

1. **Simulate** (`simulate --jobs 10`; ~ 55 min on 10 cores)
 - Seed key: (`pilot`, `poisson`, $n \cdot 10^5 + \text{rep}$, `PATTERN`)
 - Validation key: (`pilot`, `poisson`, $i < 10^5$); n from `PARAMS`, points from `PATTERN`
 - One `float32` file per n (12000×512), atomic, resumable; 3.6 GB total
2. **Fit** (`fit`)
 - Per n , fit batch: pointwise mean, s.d., Monte Carlo s.e.s
 - Weighted least squares per radius: $n m_0$, $n s_0$ on spline basis in $\log n$
 - Per n , calibration batch: max statistic under *fitted* moments $\rightarrow$ raw c_{95} , s.e.
 - Weighted spline fit of c_{95} in $\log n$
 - Store knots + coefficients (100 KB)
3. **Validate** (`validate`): rejection rate on validation patterns, overall and in 6 $\log n$ bands
4. **Use**: `Tables().statistic(curves, n)` (test), `Tables().delta_tilde(L_minus_r, nbar)` ($\tilde{\delta}$)

Results

Check	Result	Reference
Fitted m_0 vs raw, mean z^2	0.94	≈ 0.92 (1 – params/grid)
Fitted s_0 vs raw, mean z^2	0.97	≈ 0.92
Max relative error of s_0	3.9%	—
c_{95} range; max s.e.	2.75–3.21; 0.03	—
Size, 50 000 patterns	0.0515 ($z = 1.5$)	0.05
Size by n band	0.049–0.054, all $ z \leq 1.8$	0.05
Centring gap $\max m_0 /s_0/c_{95}$	0.02–0.03	—

Estimator checks

- `tests/test_lfunction.py`:
 - Weights vs brute-force circle fraction: $|\Delta| < 10^{-3}$
 - Two points at distance 0.1: $\hat{K} = 0$ below, $= 1$ above
 - $\mathbb{E}\hat{K} = \pi r^2$ under binomial (400 patterns, within 3%)
- One-off: agrees with the previous implementation (`baselines/vihrs.py`) to 10^{-9}

3 Statistical notes

Sample splitting

- Three disjoint sets: fit (m_0 , s_0), calibration (c_{95}), validation (size)
- c_{95} from same patterns as moments $\Rightarrow$ downward-biased critical value
- Size on calibration batch is 5% by construction $\Rightarrow$ validation needs fresh patterns
- c_{95} computed under the *fitted* moments $\Rightarrow$ calibrates the statistic actually used
- Validation n random integers $\Rightarrow$ mostly off-grid, tests the smoother between grid points

Fit diagnostics

- $z = (\text{raw} - \text{fit})/\text{s.e.}$; pure noise $\Rightarrow$ mean $z^2 \approx 1 - \text{params}/\text{grid points}$
- s.e. of s_0 uses the fourth moment: $\sqrt{(\mu_4 - s^4)/N}/(2s)$ (small- r tails non-Gaussian)
- c_{95} s.e. from order statistics: $(q_{.96} - q_{.94})/0.02 \cdot \sqrt{.95 \cdot .05/N}$
- Per- n mean z^2 varies widely (0.07–4.5): residuals strongly correlated across $r \Rightarrow$ few effective d.o.f. per n ; band averages ≈ 1 , no trend

Centring

- $\hat{K}$ unbiased under binomial; $\hat{L} = \sqrt{\hat{K}/\pi}$ is not (Jensen) $\Rightarrow m_0 \neq 0$
- $\tilde{\delta}$ uncentred, test centred $\Rightarrow$ gap ≤ 0.03 in $\tilde{\delta}$ units on $R(n)$
- Old tables ($r \geq 0.01$ for all n): gap 0.16 at $n = 50$ — caused by low-pair radii

Interpretation of $\tilde{\delta}$

- Defined through a classical L-test $\Rightarrow$ yardstick of that test, not an omnibus distance
- Classical baseline partly calibrated to the axis: state explicitly
- Sup over r : one radius decides; multi-scale models (nested Thomas) summarised by their most visible scale
- L only: hard cores below $r_{\min}$ are invisible to $\tilde{\delta}$ (Matérn II, above)
- Design $\tilde{\delta}$ at $\bar{n}$, not realised n : per-pattern difficulty scatters around the label (most for Cox processes)
- $\tilde{\delta} = 1$ at $\bar{n}$ does not mean $S = 1$ on a pattern: S is random around $\tilde{\delta}$

Scope and limits

- Tables valid for $n \in [20, 2000]$; outside raises (simulated pool: $n = 44\text{--}961$)
- W fixed unit square; weight formula requires $r \leq 0.5$
- $R(n)$ depends on n : at small n small-scale structure invisible by construction
- Curves stored $\Rightarrow p, \alpha$, knots changeable without re-simulation